Journal of Cleaner Production 65 (2014) 97–107

Contents lists available at ScienceDirect

Journal of Cleaner Production

journal homepage: www.elsevier.com/locate/jclepro

Energy efficiency analysis on Chinese industrial sectors: an improved Super-SBM model with undesirable outputs

Hong Li [a,*], Jin-feng Shi [b]

[a] School of Economics, Peking University, Beijing 100871, China
[b] School of Management, Shanxi University, Taiyuan, Shanxi 030006, China

ARTICLE INFO

Article history:
Received 28 April 2013
Received in revised form
3 September 2013
Accepted 22 September 2013
Available online 30 September 2013

Keywords:
Energy efficiency
Industrial sectors
Undesirable outputs
Super-SBM model

ABSTRACT

In this article we proposed an improved Super-SBM model dealing with undesirable outputs under the weak disposability assumption of undesirable outputs. Energy efficiencies of various industrial sectors in China from 2001 to 2010 are measured based on this model, and the influencing factors for energy efficiency are explored by Tobit regression model. Empirical results show that, during "The Eleventh Five-year Plan", energy efficiency of each industrial sector and category has been improved to various extents, but overall efficiency variations among industries have not taken on a convergence trend. Light industry has achieved the highest energy efficiency, followed by heavy industry; while the energy efficiency of the latter has a faster growth rate compared with that of light industry; the gap between these two industries' energy efficiency has been reduced. Energy efficiency variation presents an obvious feature of industrial economy transformation. The analysis of influencing factors show that enterprise scale, industry concentration, industrial property rights structure, and government regulation all affect energy efficiency apparently, but their effects vary across industries. Lastly, based on research results, this paper gives some policy recommendations on improving energy efficiency of the industrial sectors in China.

1. Introduction

Industry dominates national economy, and industrialization is the core and foundation of economic modernization. Since the reform and opening-up policy went into force 30 years ago, industrialization has been advanced quickly in China and gained many great achievements. Thereinto, industrial product outputs rose to NO.1 of the world already and manufacture outputs account for 20% globally now, which makes China the biggest manufacturing country. Meanwhile, industry is the main source of energy and resource consumption and pollutants in the country, and impedes sustainable development seriously. The Chinese government has adopted such policies as upgrading industrial structure and accelerating technical progress. To improve energy efficiency is considered as the basic principle in realizing energy-saving and emission-cutting. Hence, measuring and analyzing the influencing factors of industrial energy efficiency is the foundation to effectively boost industrial energy efficiency and build an industrial system with low-energy consumption, low-pollution and low-emission.

Energy is important in our economic life and related with many aspects such as economy, financial markets and social stability (Hamilton, 1983; Lardic and Mignon, 2008; Song et al., 2012, 2013; Yang et al., 2013). Energy efficiency has been studied for a long time. Based on the classical definition given by World Energy Council in 1995 and Patterson (1996): energy efficiency means using less energy to produce at least equal number of services or useful outputs. Afterward, Hu and Wang (2006) proposed two methods to measure energy efficiency based on the number of elements that impact energy efficiency during manufacturing process, i.e. single factor energy efficiency and total factor energy efficiency. At present, index decomposition analysis is used to measure single factor energy efficiency, usually including Index decomposition method proposed by Sun (1998) and Fisher Index decomposition method proposed by Ang et al. (2004). As for measuring total factor energy efficiency, methods mostly used include parametric method based on stochastic frontier analysis (SFA) and non-parametric method based on data envelop analysis (DEA). Numerous researchers have been studying industrial energy efficiency. For instance, Jenne and Cattell (1983) examined the

* Corresponding author. Tel.: +86 10 6275 5658; fax: +86 10 6275 1460.
E-mail address: hlipku0201@gmail.com (H. Li).

0959-6526/$ – see front matter
http://dx.doi.org/10.1016/j.jclepro.2013.09.035

change in the ratio of energy consumed to industrial production from 1968 to 1980, and described in detail the implications of the two recessions for efficiency and industrial structure. Phylipsen et al. (1997) identified structural differences in energy intensive industries and analyzed the ways to incorporate these differences in international comparisons of energy efficiency. They concluded that structural differences could be taken into account in cross-country comparisons of energy efficiency with appropriate physical energy efficiency indicators. DeCanio (1993) studied the relationship between energy efficiency and industrial investment and found that many investments in energy efficiency fail to be made despite their apparent profitability. Bunse et al. (2011) presented a range of methods for measuring and evaluating energy efficiency improvements in manufacturing processes such as Key Performance Indicators (KPI) and Balanced Scorecard (BSC). Oda et al. (2007) compared energy efficiency across countries in power, steel, and cement sectors, and obtained the findings that new installation and continual maintenance were essential for energy efficiency.

In recent years, China's industrial energy efficiency has become a hot research topic. Andrews-Speed (2009) examined China's vigorous programs in reversing the trend of soaring national energy intensity and reducing the intensity by 20% over the period 2006–2010, and thus to evaluate the likelihood that policies of today will yield improvements over a longer period. It is suggested that China should make greater efforts to address a number of deficiencies in the following aspects: the reluctance to use economic and financial instruments; the dependency of energy policy on industrial and social policies; the nature of political decision-making and of public administration; a shortage of skills and social attitudes to energy. Chen and Yeh (1998) employed data envelopment analysis to compare energy utilization efficiency between mainland China and Taiwan from 2002 to 2007, and they found that Taiwan is better in terms of energy utilization efficiency and the Eastern region of China had higher efficiency than the Western region. Zhou et al. (2010) summarized and analyzed the energy efficiency policies in various industries from 1970 to 2010, and provided an assessment of these policies and programs for understanding issues that will play a critical role in China's energy and economic future. Hasanbeigi et al. (2010) surveyed 16 cement plants with New Suspension Preheater and pre-calciner (NSP) kiln and compared the plant energy consumption of both domestic (Chinese) and international best practice. They used the Benchmarking and Energy Saving Tool for Cement (BEST-Cement) and the bottom-up Electricity Conservation Supply Curve (ECSC) model to estimate the potential for the 16 studied cement plants in 2008. Hu (2012) examined the origin and development of energy conservation assessment (ECA) in China, and found that ECA has a great potential in energy efficiency improvement and GHGs reduction. Wang et al. (2012) evaluated energy efficiency by DEA model under the framework of total factor energy efficiency for industrial sectors of China. They found regional energy efficiency disparity in China is prominent because of technological differences, and large scale investment should be suspended in the country, especially in central and western regions.

From social, economical and biological perspectives, underperforming industrial sectors are actually gambling the odds in a public goods game. Neglecting efficiency, the industrial sectors gain some profit while in terms of the public goods in China, and there is a loss derived from individual incentives (of the firms in a given sector) of not using more efficient procedures. Many theoretical models on the public goods game have been employed to address such dilemmas in an agent-principal setting and explore the factors that contribute to such an unfavorable outcome (see Perc and Szolnoki (2010) and Perc et al. (2013) for a comprehensive understanding on coevolutionary games and group interactions). In this paper, in order to fully reflect the situations and influencing factors of industrial energy efficiency in China, we propose an improved Super-SBM model dealing with undesirable outputs[1] under the weak disposability assumption of undesirable outputs. By this efficiency model, we will measure energy efficiency of the industrial sectors from 2001 to 2010 and explore the influencing factors of energy efficiency.

The rest of the paper is organized as follows. In Section 2 we propose the improved Supper-SBM model dealing with undesirable outputs under the weak disposability assumption, and illustrate the Tobit regression method. In Section 3 we present the industrial sectors and its categories, input–output data and the analysis results of energy efficiency in Chinese industrial sectors from 2001 to 2010. We explore influencing factors of the energy efficiency by Tobit regression model in Section 4 and give the conclusions in Section 5.

2. Methodology

2.1. Improved SBM and Super-SBM models

Tone (2001) proposed a slacks-based measure (SBM) of efficiency. Unlike traditional DEA model, the slack variables in SBM model are directly added into the target function. The SBM method is thus non-radial and deals with input/output slacks directly, eliminating the radial and oriented deviation (Song et al., 2013). In addition, the economic interpretation of the evaluation model is to make actual profit maximization rather than simply maximize efficiency ratio as in CCR or BCC model. As known to all, undesirable outputs such as waste water, exhausted gas, are unavoidable for production and living, so it is necessary to take account the undesirable outputs into efficiency evaluation model (Seiford and Zhu, 2002). Tone and Sahoo (2003) proposed a new measuring efficiency scheme with undesirable outputs based on SBM model. In order to better handle the undesirable outputs, we should categorize disposability assumptions, that is, strong disposability or weak disposability. As pointed out in Färe and Grosskopf (2004) and Ray (2004), the strong disposability states that undesirable outputs can be discarded at no cost which undermines the usefulness of the concept; while weak disposability assumes that the outputs are weakly disposable while only the sub-vector of the desirable outputs is strongly disposable. In this paper, we will combine the assumption of weak disposability with the SBM model as well as Super-SBM model, and thus obtain an improved efficiency model considering the undesirable outputs.

Now we consider a production system with n DMUs, each unit has three factors: inputs, desirable outputs and undesirable outputs (environment pollution, such as CO_2, SO_2, etc), as represented by three vectors: $x \in R^m$, $y^g \in R^{s_1}$, $y^b \in R^{s_2}$. We define the matrices X, Y^g, Y^b as follows:

$$X = [x_1, x_2, \ldots, x_n] \in R^{m \times n}, Y^g = [y_1^g, y_2^g, \cdots, y_n^g] \in R^{s_1 \times n}, Y^b = \left[y_1^b, y_2^b, \ldots, y_n^b\right] \in R^{s_2 \times n},$$

[1] In accordance with the global environmental conservation awareness, undesirable outputs of productions and social activities, e.g., air pollutants and hazardous wastes, are being increasingly recognized as dangerous and undesirable. Thus, development of technologies with less undesirable outputs is an important subject of concern in every area of production. Traditional efficiency measurement model usually assumes that producing more outputs relative to less input resources is a criterion of efficiency. In the presence of undesirable outputs, however, technologies with more good (desirable) outputs and less bad (undesirable) outputs relative to less input resources should be recognized as efficient.

and assume that $X > 0$, $Y^g > 0$, $Y^b > 0$. Then the production possibility set (P) is defined by assuming

$$P = \left\{ \left(x, y^g, y^b\right) \middle| x \geq X\lambda, y^g \leq Y^g\lambda, y^b = Y^b\lambda, \lambda \geq 0 \right\},$$

where λ is the intensity vector. According to the SBM model in Tone (2003) and weak disposability assumption, the improved SBM model dealing with undesirable outputs for evaluating DMU (x_0, y_0^g, y_0^b) is as follows.

$$[\text{Improved SBM}] \quad \rho^* = \min \frac{1 - \frac{1}{m}\sum_{i=1}^{m} \frac{S_i^-}{x_{i0}}}{1 + \frac{1}{s_1}\sum_{r=1}^{s_1} \frac{S_r^g}{y_{r0}^g}}$$

Subject to

$$\begin{aligned} x_0 &= X\lambda + S^- \\ y_0^g &= Y^g\lambda - S^g \\ y_0^b &= Y^b\lambda \\ S^- &\geq 0, S^g \geq 0, \lambda \geq 0 \end{aligned} \tag{1}$$

where $S = (S^-, S^g)$ corresponds to the slacks in inputs and desirable outputs. The optimization function value of ρ^* is the efficiency value of the Decision Making Unit (x_0, y_0^g, y_0^b). m, s_1 and s_2 stand for the number of factors for inputs, desirable outputs and undesirable outputs. By Charnes–Cooper transformation, we can transform the above nonlinear program into a linear program in the following equivalent form.

$$[\text{Improved SBM}'] \quad \tau^* = \min t - \frac{1}{m}\sum_{i=1}^{m} \frac{S_i^-}{x_{i0}}$$

Subject to

$$\begin{aligned} 1 &= t + \frac{1}{s_1}\sum_{r=1}^{s_1} \frac{S_r^g}{y_{r0}^g} \\ x_0 t &= X\Lambda + S^- \\ y_0^g t &= Y^g\Lambda - S^g \\ y_0^b t &= Y^b\Lambda \\ S^- &\geq 0, S^g \geq 0, \Lambda \geq 0, t > 0 \end{aligned} \tag{2}$$

However, most empirical results of the efficiency evaluation research have a common phenomenon, that is, plural Decision Making Units have the "efficient status" denoted by 100%. So how to rationally discriminate between these efficient DMUs is important for efficiency ranking and influence factors analysis. Based on the above improved SBM model, we improve the Super-SBM model in Tone (2002) under the assumption of weak disposability. The improved Super-SBM model dealing with undesirable outputs which is used for evaluating the SBM-efficient DMUs is as follows:

$$[\text{Improved Super-SBM}] \quad \delta^* = \min \frac{\frac{1}{m}\sum_{i=1}^{m} \frac{\bar{x}_i}{x_{i0}}}{\frac{1}{s_1}\sum_{r=1}^{s_1} \frac{\bar{y}_r^g}{y_{r0}^g}}$$

Subject to

Table 1
Names and details of 36 sub-industries.

Industry classification	Industry code	Industry name
Mining industry (I)	SER 01	Coal mining and washing
	SER 02	Oil and natural gas mining
	SER 03	Ferrous metal mining
	SER 04	Non-ferrous metal mining
	SER 05	Non-metal mining
Light industry (II)	SER 06	Agricultural products processing
	SER 07	Food manufacturing
	SER 08	Beverage manufacturing
	SER 09	Tobacco manufacturing
	SER 10	Textile industry
	SER 11	Textile clothes, shoes, hats manufacturing
	SER 12	Leather, fur, feather manufacturing
	SER 13	Wood processing, and wood, bamboo, cane, palm, and straw manufacturing
	SER 14	Furniture manufacturing
	SER 15	Papermaking and paper products
	SER 16	Press and intermediary replication
	SER 17	Cultural, educational and sports goods manufacturing
Heavy industry (III)	SER 18	Oil processing, coking and nuclear fuels processing
	SER 19	Manufacturing of chemical materials and products
	SER 20	Manufacturing of medicines
	SER 21	Manufacturing of chemical fiber
	SER 22	Manufacturing of rubber
	SER 23	Manufacturing of plastics
	SER 24	Manufacturing of non-metal products
	SER 25	Smelting and rolling process of non-ferrous metal
	SER 26	Smelting and rolling process of Ferrous metal
	SER 27	Manufacturing of metal products
	SER 28	Manufacturing of ordinary machinery
	SER 29	Manufacturing of special equipments
	SER 30	Manufacturing of transportation and equipments
	SER 31	Manufacturing of electric machines
	SER 32	Manufacturing of communication device, computers and other electronic equipments
	SER 33	Manufacturing of instruments, cultural and official mechanics
Electricity, gas and water industry (IV)	SER 34	Production and supply of electricity, power
	SER 35	Gas production and supply
	SER 36	Water production and supply

$$\begin{aligned} \bar{x} &\geq \sum_{j=1,\neq 0}^{n} \lambda_j x_j \\ \bar{y}^g &\leq \sum_{j=1,\neq 0}^{n} \lambda_j y_j^g \\ \bar{y}^b &= \sum_{j=1,\neq 0}^{n} \lambda_j y_j^b \\ \bar{x} &\geq x_0, \bar{y}^g \leq y_0^g, \bar{y}^b \leq y_0^b, \bar{y}^g \geq 0, \lambda \geq 0 \end{aligned} \tag{3}$$

It is worth noting that the above improved SBM model and the Super-SBM model dealing with undesirable outputs are under the assumption of constant returns-to-scale (CRS). Also we also can relax and extend these models to the variable returns-to-scale (VRS) case with the restrictions $\sum_{i=1}^{n} \lambda_i = 1$ in model (1) and $\sum_{i=1,\neq 0}^{n} \lambda_i = 1$ in model (3) respectively.

2.2. Tobit regression model

Although the improved efficiency model dealing with undesirable outputs can contribute to the improvement of the efficiency performance, it does not capture the key factors affecting the energy efficiency. Nevertheless, we could use multivariate analysis to explore the influencing factors. Because the efficiency value is non-negative, the general estimation method as Ordinary Least Square (OLS) will lead to biased and inconsistent estimating results. Tobit regression model is based on the principle of maximum likelihood estimation to get the consistent parameter estimation. This regression model belongs to the limited dependent variable or truncation econometrics model, and the dependent variable can only be observed in a restricted way. The standard Tobit regression model is as follows:

$$y_i^* = \beta X_i + \mu_i \sim N\left(0, \sigma^2\right), \; i = 1, 2, \ldots, n, \tag{4}$$

where i stands for the ith DMU, y_i^* a latent (i.e. unobservable) variable, X_i is $K \times 1$ matrix on behalf of independent variables. μ is stochastic error and submits to $N(0,\sigma^2)$. The limited value y_i is:

$$y_i = \begin{cases} y_i^*, & y_i^* > 0 \\ 0, & y_i^* \le 0 \end{cases}.$$

3. Data and empirical results

3.1. Data and industrial sectors

Currently, existing researches focus on energy efficiency of one single sector and disregard the undesirable outputs such as environment pollution. This paper is more concerned with distribution characteristics and the evolving trend of energy efficiency in the whole industry of China. Hence, we select 36 sub-industries in China from 2001 to 2010 and classify them into 4 categories, i.e. mining industry, light industry, heavy industry, and electricity, gas and water industry (see Table 1). Subsequently, we can not only analyze the characteristics and disparities of energy efficiency across different sub-industries, but also study the energy efficiencies, evolving trends and influencing factors of different industrial categories. In addition, the selected period spans two "Five-Year Plans",[2] i.e. "The Fifth Five-Year Plan" and "The Eleventh Five-Year Plan". So we can have a deeper understanding of the variation of energy efficiency of different industries during economic growth and the relationship between the formation and implementation of energy-saving and emission-reducing polices.

In researches on energy efficiency, input indicators generally include capital, labor, and energy consumption. In this paper, we utilize the outstanding net value of fixed asset of the enterprises above designated scale as the proxy for capital input; average amounts of total employees of the enterprises above designated scale as the proxy for labor input; total energy consumption of sub-industries as the proxy for energy input. Basically, it is recognized that DMU is relatively more effective during manufacturing process when the ratio of input over output is as little as possible. However, besides the desirable outputs, undesirable outputs such as environment pollutants are generated during manufacturing process as well. Hence, we divide outputs into desirable outputs and

Table 2
Descriptive statistical characteristics of input and output variables.

Variable	Inputs			Outputs	
	Capitalx_1	Laborx_2	Energy x_3	Industrial output y^g	Wastes y^b
Mean	3.0825	1.8707	4.4717	9.0646	7.4405
Median	1.4892	1.2415	1.3226	4.8643	0.3190
Maximum	47.9014	7.7275	56.4130	55.4526	1291.2890
Minimum	0.1110	0.1400	0.0955	0.1849	0.0033
Std. Dev	5.1395	1.6177	8.3408	10.6865	72.8702
Skewness	5.0705	1.1391	3.4669	1.9756	15.8621
Kurtosis	35.9951	3.5349	17.2509	6.7711	272.3866

Note: Each sample has 360 observations, for the panel data includes 5 indicators of 36 industrial sectors from 2001 to 2010.

undesirable outputs. We define the gross output value of the enterprises above designated scale as desirable output, and we proxy the total emission of the three wastes (waste gas, waste water, industrial residue) in all sub-industries for undesirable output because it lacks accurate data of pollutants emission like CO_2, SO_2, NO_2 in China. Data is cited from China Statistical Yearbook and China Energy Statistical Yearbook published by the National Bureau of Statistics of China. The descriptive statistical characteristics of above mentioned input/output variables are shown in Table 2.

Table 2 suggests that median of different indicators is much smaller than the mean value, and a larger standard deviation shows unbalanced production status of different industrial sectors, which is more prominent in terms of desirable outputs and undesirable outputs. Moreover, the maximum can be as much as more than 500 times of the minimum for energy consumption and outputs in different industrial sectors and both are closely related to industry attributes. Therefore, an analysis in-depth of the energy efficiency in industrial sectors of China and the exploration of influencing factors are helpful for the implementation of energy conservation and emission reduction policies during the "Twelfth Five-Years Plan". Although not all industrial sectors are energy-intensive, we find that correlation between energy consumption and desirable outputs is still up to 0.5604 from the Pearson coefficient in Table 3. In other words, the DMUs' production process has some so-called "isotonicity". In addition, the correlation between inputs and undesirable outputs is weak and insignificant, in line with the actual production expectation. Therefore, the energy efficiency measured by the improved Super-SBM model is reliable, and the research results are completely believable.

3.2. Energy efficiency performance in China's industrial sectors

In this section, we will measure the energy efficiency of Chinese industrial sectors from 2001 to 2010 by the improved Super-SBM model, and analyze the energy efficiency performance and development trends in different industrial sectors.

3.2.1. Energy efficiency characteristics of Chinese industrial sectors in 2001–2010

Fig. 1 presents that Tobacco manufacturing (SER 09), Leather, fur, feather, manufacturing (SER 12) and Manufacturing of electric machines (SER 31) share the highest energy efficiency, but the energy efficiency in Manufacturing of electric machines (SER 31) is above 0.5 almost every year. The energy efficiencies of agricultural products processing (SER 06), agricultural products processing (SER 11), furniture manufacturing (SER 14), oil processing, coking and nuclear fuels processing (SER 18), communication device, computers and other electronic equipments (SER 32) and manufacturing of instruments, cultural and official mechanics (SER 33) are higher than the average efficiency value. However, coal mining and washing (SER 01), oil and natural gas mining (SER 02),

[2] The five-year plans of People's Republic of China (PRC) are a series of social and economic development initiatives. The economy was shaped by the Communist Party of China (CPC) through the plenary sessions of the Central Committee and national congresses. The party plays a leading role in establishing the foundations and principles of Chinese communism, mapping strategies for economic development, setting growth targets, and launching reforms. Planning is a key characteristic of centralized, communist economies, and one plan established for the entire country normally contains detailed economic development guidelines for all its regions. In order to more accurately reflect China's transition from a Soviet-style planned economy to a socialist market economy (socialism with Chinese characteristics), the name of the 11th five-year program was changed to "guideline".

Table 3
Pearson coefficients between input and output variables.

Indicator	x_1	x_2	x_3	Indicator	x_1	x_2	x_3
y^g	0.6295*** (12.4851)	0.7453*** (21.1482)	0.5604*** (12.8038)	y^b	0.1361*** (2.5991)	0.0438 (0.8301)	0.1020* (0.0532)

Note:“*”,“**”,“***” present their significance respectively at levels of 10%, 5% and 1%.

production and supply of electricity, power (SER 34), gas production and supply (SER 35) and water production and supply (SER 36) have the lowest energy efficiency and high energy consumption. On the whole, energy efficiencies of most industrial sectors are relatively lower, including and efficiency values ranges from 10% to 50%, which suggests a big gap compared with western developed countries.

Fig. 2 shows that various sectors (except water production and supply) of mining industry (I) and electricity, gas and water industry (IV) present the rising energy efficiency during 2001–2010, and efficiency disparity among sectors has been not significantly decreased. Non-ferrous metal mining (SER 04) and ferrous metal mining (SER 03) of mining industry (I) have the highest energy efficiency, whereas coal mining and washing (SER 01) gain the lowest efficiency. Production and supply of electricity, power (SER 34), gas production and supply (SER 35) of electricity, gas and water industry (IV) almost share the same level of energy efficiency, but the energy efficiency of water production and supply (SER 36) is relatively low and shows no variation during these years.

Fig. 3 presents that energy efficiencies of various sectors in light industry (II) have increased during 2001–2010 and efficiency growth rates remain quite stable. Tobacco manufacturing (SER 09) and Leather, fur, feather, manufacturing (SER 12) have relatively higher energy efficiency, and energy efficiencies of textile industry (SER 10), papermaking and paper products (SER 15) and press and intermediary replication (SER 16) are lower. From Fig. 4, we learn that except for manufacturing of non-metal products (SER 24), energy efficiencies of various sectors in heavy industry (III) have the same variation tendency as those in other industrial categories. In addition, oil processing, coking and nuclear fuels processing (SER 18) and manufacturing of electric machines (SER 31) have higher energy efficiency than that of other sectors in light industry, while manufacturing of non-metal products (SER 24) has lower energy efficiency.

3.2.2. Energy efficiency of the four major industrial categories and their tendency analysis

According to the energy efficiency of each industrial sector and industry classification in Table 1, we can give the efficiency of the four major industrial categories in China from 2001 to 2010 and their evolving tendency in Table 4 and Fig. 5.

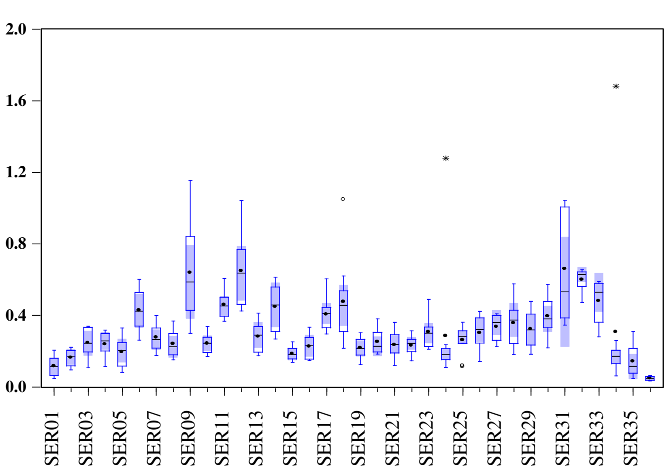

Fig. 1. Boxplot of energy efficiency of industrial sectors in 2001–2010 in China.

Table 4 shows that, the annual average energy efficiency values of mining industry(I), light industry(II), heavy industry (III) and electricity, gas and water industry (IV) during 2001–2010 are 0.1928, 0.3744, 0.3584, 0.1674 respectively. It indicates that under the current conditions, energy efficiency of mining, electricity, gas and water industry can be upgraded by at least 15% to catch up with the annual average energy efficiency of light and heavy industry. Fig. 5 shows that during 2001–2010, light industry obtains the highest energy efficiency among the 4 industrial categories, followed by heavy industry. Energy efficiency of heavy industry is improved more than the light industry, and the energy efficiency gap between them is narrowed due to multiple factors such as industry attributes, energy structures and policies orientations. In real manufacturing process, light industry depends less on resources and develops intensively. Meanwhile, the technology and equipment has approached closely to world leading level. While heavy and mining industries as energy-intensive industries, mainly rely on infrastructure resources like coals, and have extensive development mode and suffer laggard technology equipment. During “The Eleventh Five-Year Plan”, the government proposed policies to transform the economic development structure as well as to cut down energy consumption and reduce emissions. For example, the related policies included integrating the mining industry of Shanxi; sweeping out laggard production capacity of iron and steel industry; shutting down enterprises producing large amount of pollution. Hence, energy efficiencies of different industries in China have been promoted to various degrees. Heavy industry has transformed from extensive pattern to intensive pattern, and attained higher energy utilization efficiency above 0.5 in 2010. By contrast, energy efficiencies of mining industry and electricity, gas, and water industry are relatively lower, and achieve no improvement compared with light and heavy industry. Finally, the energy efficiency of the entire industry in China is not satisfactorily high, which needs a further improvement.

3.2.3. Comparison of energy efficiency across industrial sectors in the two “Five Year Plans”

“The Five Year Plan” is an important component of the long-term development in terms of the national economy, and it mainly concentrates on the planning of national construction, distribution of productivity and the proportion of national economy. Therefore, we will use the developing programs of “The Tenth-Five Plan” and “The Eleventh-Five Plan” to make further analysis on the energy efficiency characteristics in China. The energy efficiency differences in different industrial sectors between the two periods can be discerned in Fig. 6.

Fig. 6 illustrates that during “The Tenth-Five Plan” and “The Eleventh-Five Plan”, except for manufacturing of non-metal products (SER 24) and manufacturing of electric machines (SER 31), every industrial sector has shown a significant improvement in energy efficiency. This attributes to the fact that “The Tenth-Five Plan” gave priority to development and preliminarily established a market-oriented economic system, and the transformation of economic growth pattern was rather low with a growing need of energy consumption. However, “The Eleventh-Five Plan” clearly stipulated that energy consumption per GDP should be lowered by

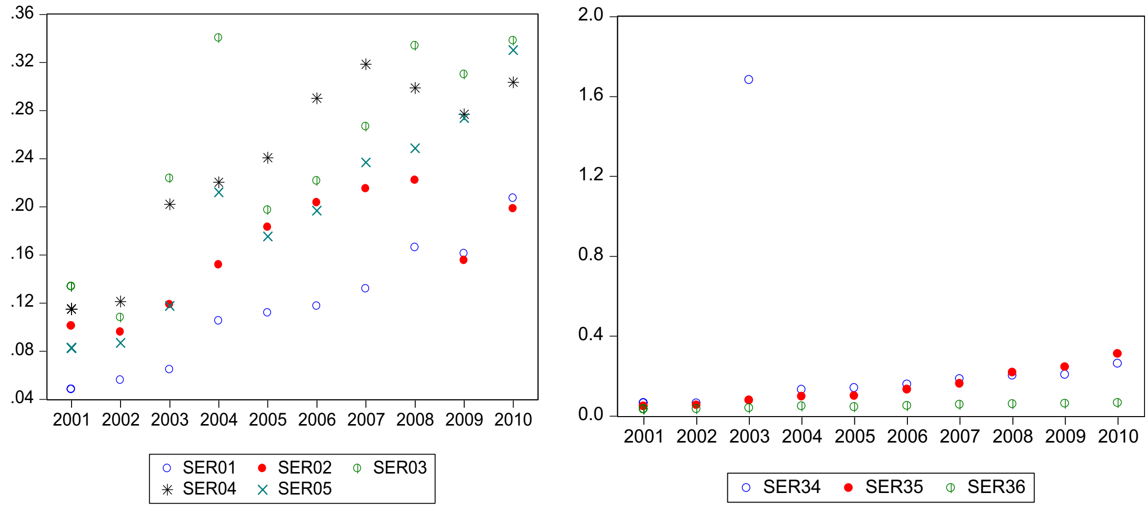

Fig. 2. Energy efficiency tendency of extractive industries and electricity, gas and water industries in China in 2001–2010.

around 20%, and particularly incorporated energy-saving and consumption reducing index into the five-year plan for the first time. "The Eleventh-Five Plan" required the implementation of the basic national resources conservation and environment protection policies. That means the establishment of the recyclable and sustainable national economic system with low input, high output, low energy consumption and low emissions, and the construction of a resource-saving and environment-friendly society.

On the whole, industrial energy efficiency variations in China are characterized with three features: (a) term-end (2010) energy efficiency of 36 sub-industries is higher than that of term-beginning (2001). The entire industrial energy efficiency rises from 0.19 to 0.47, indicating an overall improvement of energy utilization efficiency. These results are in accordance with the targets of "The Eleventh-Five Plan" to reinforce energy-saving and emission-cutting and reduce per GDP energy consumption. (b) Energy efficiency variations among industries have not shown any obvious convergence characteristics. Though almost every individual industry has achieved a higher energy efficiency value, the standard variation has been increased which might result from industry structure adjustment and strategy transformation. (c) Light industry has a better performance in the context of energy

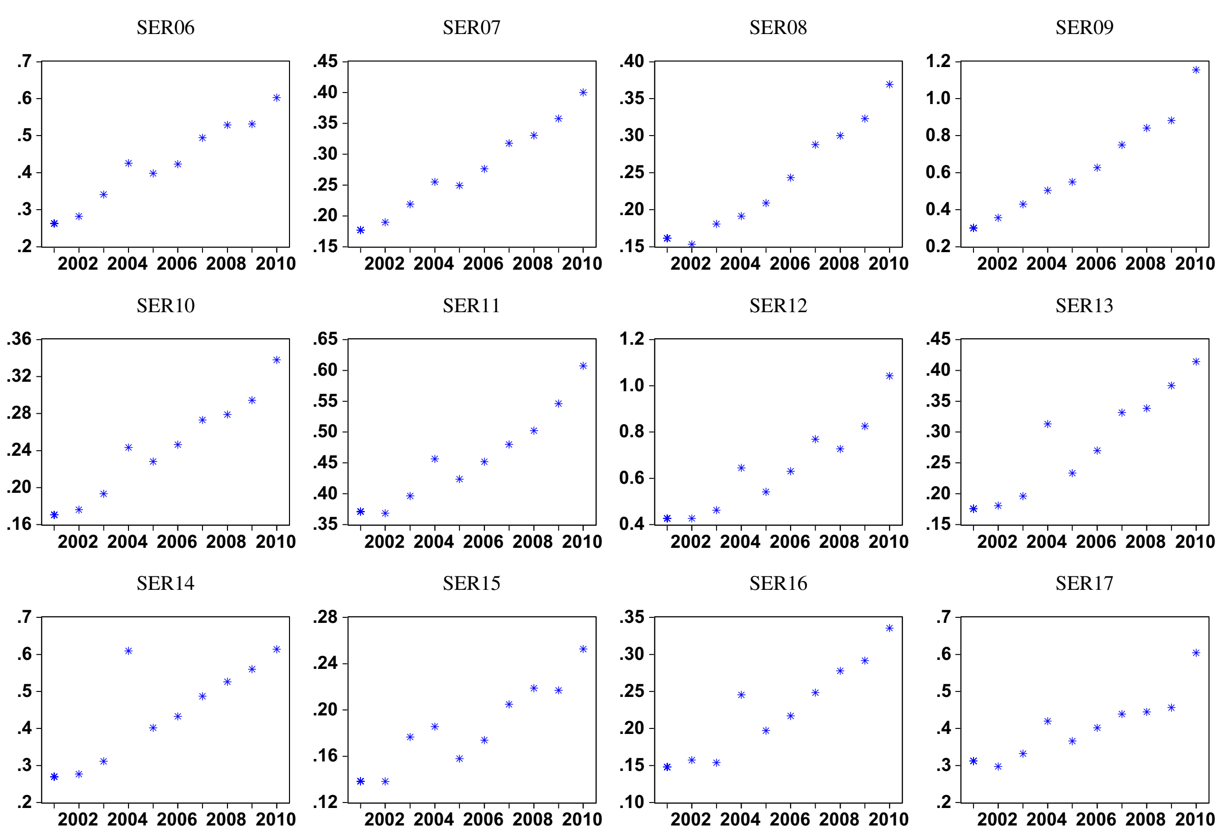

Fig. 3. Energy efficiency tendency of light industries in 2001–2010 in China.

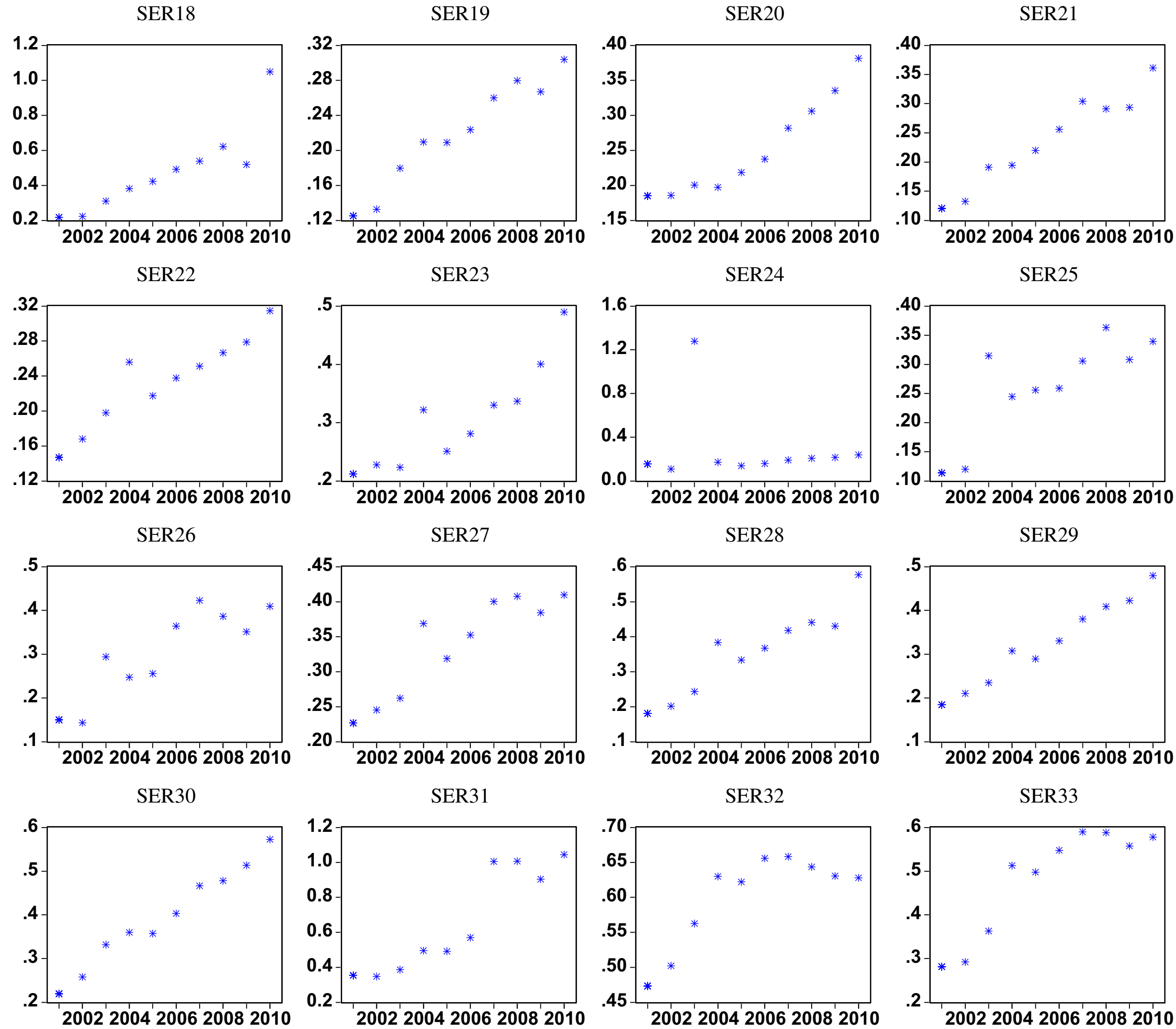

Fig. 4. Energy efficiency tendency of heavy industries in 2001–2010 in China.

efficiency, while heavy industry has a higher growth rate of energy efficiency. Nonetheless, the whole industry in China is still of relative low energy efficiency and demanding a further improvement.

4. Influencing factors of energy efficiency in Chinese industrial sectors

In the above study, we measure the energy efficiency of industrial sectors in China from 2001 to 2010, and analyze the high energy efficiency disparity across various industries and some evolving trends. In order to further study the causes for the energy efficiency disparity, we will use the Tobit regression model to explore relevant influencing factors on energy efficiency from such aspects as industry structure, energy consumption, technology innovation and government regulation. Meanwhile, we make some pre-judgments for each factor.

4.1. Influencing factors' selection

"The Tenth-Five Plan" and "The Eleventh-Five Plan" issued the hot topic that whether industrial economy transformation could enhance of energy efficiency. Meanwhile, the industrial economy of China is basically in the stage of strengthening market competitiveness, boosting the reform of state-owned enterprises, and perfecting the legal structure of property rights. For this purpose, from the perspective of industry structure, we employ the industry scale and industry intensiveness as proxy for market structure,

Table 4
Energy efficiency of Chinese four major industrial categories in 2001–2010.

Year	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010
I	0.0961	0.0936	0.1454	0.2060	0.1817	0.2060	0.2339	0.2540	0.2355	0.2756
II	0.2427	0.2500	0.2825	0.3743	0.3295	0.3659	0.4233	0.4428	0.4715	0.5613
III	0.2090	0.2186	0.3484	0.3301	0.3187	0.3584	0.4251	0.4395	0.4256	0.5110
IV	0.0487	0.0507	0.5993	0.0920	0.0943	0.1129	0.1340	0.1591	0.1708	0.2120
O	0.1912	0.1977	0.3192	0.3077	0.2846	0.3193	0.3737	0.3915	0.3933	0.4701

Note: I-Extractive Industry, II-Light Industry, III-Heavy Industry, IV-Electricity, Gas and Water Industry, O-Overall Average.

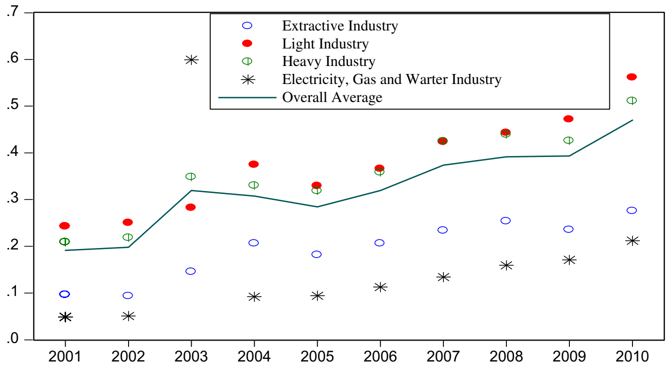

Fig. 5. Energy efficiency of four major industrial categories in China from 2001 to 2010.

reflecting the industrial economies of scale to some extent; the proportion of state-owned and state-owned holding enterprises to the gross industrial output value stands for property rights structure, and capital-labor structure for human costs. In addition, we introduce ES^2 (see Table 5) into regression model in order to analyze whether the variation of industry scale to energy efficiency has a U-type curve or inverse U-type curve. From the perspective of energy consumption, this paper also analyzes the impact of industrial energy structure adjustment on energy efficiency in terms of coal, oil and electricity. Besides, research & development investment and human capital are important support for technology improvement and innovation, so we utilize R&D investment (RDI) as proxy for the investment of searching technical progress, and R&D researchers (RDR) as proxy for the investment of R&D human capital. In view of government regulations, we introduce foreign direct investment as the indicator of the market openness of different industries and pollution control investment as the indicator of disposing costs resulted from governmental irrational behaviors during marketization process. Given the huge diversity of resources endowments, pollutant emissions, technology compositions and industrial cycle in China, we study the whole sample and industry group sample for exploring the influencing factors of energy efficiency on the level of the whole industry and different industries. The data is from China Statistical Yearbook, China Energy Statistical Yearbook and China environment Statistical Yearbook. Table 5 shows the concrete definition of variable index and corresponding symbols.

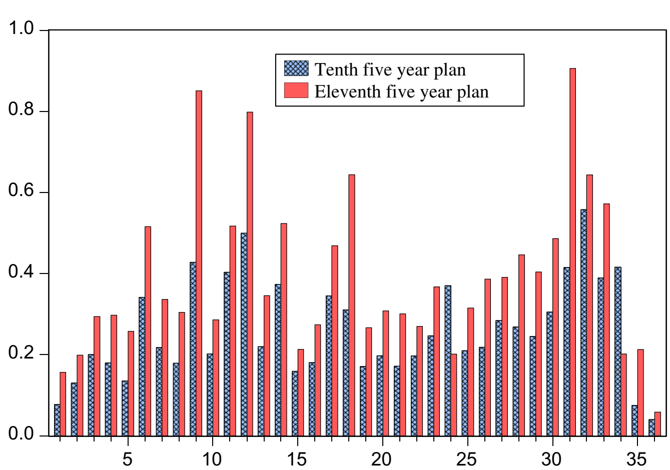

Fig. 6. Energy efficiency of industrial sectors in the two "Five Year Plans" in China.

Table 5
Influence factors and relevant symbols.

	Variables	Symbols	Variables' definition and unit	Pre-judgment
Industry structure	Enterprise scale	ES	Ratio of gross output value to enterprise amount in sub-industry (0.1 billions/per unit)	Unknown
		ES^2	Quadratic of ES	Unknown
	Industry concentration	IC	Ratio of large and medium sized enterprises' output value to sub-industry output value (%)	Positive
	Property-right structure	PS	Ratio of state-owned and state-owned holding enterprises' output value to sub-industry output value (%)	Negative
	Capital-labor structure	CL	Ratio of net value of fixed assets to employees amount (10 thousands/per person)	Unknown
Energy consump tion	Coal	MC	Ratio of coal consumption to sub-industry's energy consumption (%)	Negative
	Oil	OC	Ratio of oil consumption to sub-industry's energy consumption (%)	Negative
	Electricity	EC	Ratio of electricity consumption to sub-industry's energy consumption (%)	Positive
Technology Innovation	Research and development investment	RDI	Ratio of R&D investment to sub-industry's fixed assets (%)	Positive
	R&D researchers	Ln (RDR)	R&D researchers' logarithm	Positive
Government regulation	Pollution costs	Ln (PC)	Logarithm of total costs of controlling (including various industries' waste water and waste gas)	Positive
	Foreign domestic investment	FDI	Ratio of FDI and investment from Hongkong, Macao and Taiwan to sub-industry's output value (%)	Unknown

4.2. Tobit regression model of the energy efficiency and empirical results

Considering influencing factors, the Tobit regression model between energy efficiency (EEI) in industrial sectors and influencing factors is:

$$\begin{aligned} EEI_{it} = {} & \beta_0 + \beta_1 ES_{it} + \beta_2 ES_{it}^2 + \beta_3 IC_{it} + \beta_4 PS_{it} + \beta_5 CL_{it} + \beta_6 MC_{it} \\ & + \beta_7 OC_{it} + \beta_8 EC_{it} + \beta_9 RDI_{it} + \beta_{10} \ln(RDR)_{it} \\ & + \beta_{11} \ln(PC)_{it} + \beta_{12} FDI_{it} + \varepsilon_{it} \end{aligned} \tag{5}$$

EEI_{it} on the left means the energy efficiency of the ith industrial sector in the tth year. Symbols on the right mean the corresponding influencing factors in the i th industrial sector in the tth year. β_0, $\beta_1, \ldots \beta_{12}$ are the unknown coefficients, and ε_{it} is the random error. For considering different types of explanatory variables, we

Table 6
Tobit regression results of energy efficiency of Chinese industrial sectors.

	All sample					(I)	(II)	(III)	(IV)
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 5	Model 5	Model 5	Model 5
ES	0.0406*** (6.10)	0.0391*** (6.25)	0.0404*** (6.14)	0.0412*** (6.62)	0.0441*** (6.27)	−0.0176** (−2.28)	0.0782*** (8.16)	0.0102 (0.39)	−0.312* (−2.07)
ES^2	−0.0009*** (−5.27)	−0.0008*** (−5.41)	−0.0008*** (−5.24)	−0.0009*** (−5.69)	−0.0009*** (−5.47)	0.0003** (2.67)	−0.001*** (−4.35)	0.0078*** (3.48)	0.0189 (0.76)
IC	0.2909*** (4.57)	0.2965*** (4.78)	0.275*** (4.16)	0.2405*** (3.23)	0.1846** (2.21)	0.1896 (1.28)	0.0152 (0.21)	0.3704*** (2.71)	4.0509** (2.69)
PS	−0.4717*** (−9.22)	−0.4539*** (−9.83)	−0.452*** (−9.46)	−0.3775*** (−6.17)	−0.3400*** (−4.48)	−0.2698** (−2.61)	−0.7203*** (−5.84)	−0.3446*** (−2.99)	−3.6393** (−2.28)
CL	−0.0006 (−0.96)				−0.0008 (−1.11)	0.0019*** (2.81)	−0.0044 (−1.47)	−0.01*** (−3.82)	0.0027 (0.68)
MC		−0.0588 (−0.88)			−0.0393 (−0.54)	0.0854 (0.81)	0.499** 6(2.59)	−0.1261 (−1.04)	−0.7817 (−1.06)
OC		−0.0007 (−0.07)			−0.0039 (−0.35)	0.5505 (1.42)	2.4216*** (6.14)	−0.0017 (−0.17)	0.8522 (0.82)
EC		0.4356*** (2.71)	0.5247*** (3.88)	0.4677*** (3.41)	0.4377*** (2.49)	0.5052* (1.95)	1.563*** (2.79)	0.6577** (1.99)	−0.2241 (−0.51)
RDI			0.1109 (1.3)		0.0944 (1.11)	0.1513 (0.76)	0.63*** (2.73)	0.0553 (0.72)	1.7476 (1.38)
Ln(RDR)			−0.001 (−0.25)		−0.0009 (−0.22)	−0.0049 (−1.6)	−0.0093 (−1.57)	0.0054 (1.06)	−0.0394* (−1.75)
Ln(PC)				−0.0019 (−0.29)	0.0013 (0.17)	0.0385** (2.55)	0.0333*** (3.82)	0.0222* (1.79)	0.0752 (1.4)
FDI				0.1422* (1.94)	0.1623*** (2.01)	1.0323*** (3.23)	−0.1826 (−1.46)	−0.0714 (−0.62)	−0.9822 (−0.63)
Constant	0.257*** (8.50)	0.1685*** (2.91)	0.1454*** (3.11)	0.1345 (1.57)	0.1400 (1.27)	−0.3925* (−2.01)	−0.5295** (−2.21)	−0.0475 (−0.26)	0.1294 (0.17)
Log likelihood	114.08	121.63	110.57	123.65	113.69	83.70	102.19	78.49	1.98
LR chi2	118.6***	133.71***	126.43***	137.73***	132.66***	76.73***	160.79***	93.50***	17.72

Note: "*", "**", "***" present their significance at levels of 10%, 5% and 1% respectively.

estimate different Tobit regression models based on the maximum likelihood estimation, and the corresponding results are presented in Table 6.

4.2.1. Impacts of enterprise scale

Enterprise scale is one of the important determining factors for energy utilization efficiency. Estimation results of the whole sample in Table 6 show that enterprise scale (ES) is overall positively related with energy efficiency, and industry's energy efficiency will rise by around 4% when average enterprise scale increases by 1 point. This means that currently, China industry scale has not yet reached the optimal level. Therefore, increasing the outputs of industries to a designated scale will contribute to higher scale efficiency and improve the energy efficiency. However, as the enterprise scale expands, the complexity of industry internal structure leads to more consumption on energy and resources, thus may offset the benefits brought by scale expansion, decrease the scale efficiency and even lead to diseconomies of scale. So we take account of quadratic term of enterprise scale (ES^2) to measure whether there is a nonlinear relationship between enterprise scale and energy efficiency. Results show that quadratic term of enterprise scale formed by the whole sample has a negative sign, demonstrating that with the increase of enterprise scale, industry energy efficiency takes on inverse U-shape, i.e., first rising and then falling. In addition, the estimation results by group sample show that light industry (II) is in consensus with that of the whole sample, while enterprise scale of mining industry (I) is positively related with energy efficiency. With expansion of enterprise scale, energy efficiency of mining industry (I) and light industry (III) first drops and then rises, i.e. a characteristic of U-type curve. Hence, in order to meet the demand of economic strategic development and higher energy utilization efficiency, it is needed to prudently deal with the impacts of enterprise scale on industrial energy efficiency respectively.

4.2.2. Impacts of industry concentration

The estimation results of the whole sample in Table 6 show that, industry concentration is positively related to energy efficiency, implying that higher industry concentration is beneficial to energy efficiency. Analysis on enterprise scale indicates that energy efficiency has increasing returns to scale. High industry concentration means an industry with larger enterprise scale accordingly, and it helps improve energy efficiency. However, group tests indicate that there is no significant relationship in mining industry (I) and light industry (II). This is probably because enterprises with higher degree of industry concentration are able to attain relative cheaper resources, so they do not have the motivation to improve energy utilization efficiency.

4.2.3. Impacts of property rights structure

The estimation results of the whole sample and the group sample in Table 6 show that the relationship between property rights structure and energy efficiency is significantly positive. Increasing proportion of state-owned enterprises would reduce energy efficiency, which is in accordance with the current research conclusions. That mainly lies in the ambiguous property rights and ownership scheme as well as the rigid operational mechanism, which hinder the improvement of energy efficiency. Additionally, industries with high pollution are generally capital-intensive, and capital-intensive industries tend to have a higher state-owned ratio. Hence, to further decrease the state-owned ratio and deepen the reform of state-owned enterprises will benefit industrial energy efficiency.

4.2.4. Impacts of energy consumption structure

From the estimation results of the whole sample, electricity consumption is positively related with industrial energy efficiency at the 1% significance level. When electricity consumption proportion increases by 1%, industrial energy efficiency will increase by 43.77%. Estimation results of group tests imply that electricity consumption is positively related to industrial energy efficiency at the 10% significance level, and improvements of energy efficiency in coal, oil and electricity consumptions have a significant positive impact on energy efficiency of light industry. In addition, for heavy industry, coefficients of coal and oil consumptions are not significant, but negative. This indicates that during manufacturing process, boosting the proportion of clean energy such as electricity can reduce pollutant emissions like CO_2 and enhance the energy efficiency of the entire industry.

4.2.5. Impacts of government regulation

Estimation results in Table 6 show that investment on pollutant control has a significant positive impact on upgrading energy efficiency. Some studies show that western countries gradually transfer industries with high energy consumption and high-pollution to China, and this hinders the improvement of industrial energy efficiency in China. Moreover, some studies indicate that China should absorb advanced management experience and technologies to best leverage foreign direct investment. In order to have a better knowledge of the FDI's impact on Chinese industrial energy efficiency, we study from two perspectives, i.e. the industry as whole and industrial groups (see Table 6). Estimation results of the whole sample demonstrate that FDI benefits the improvement of industrial energy efficiency, while group tests show that the impacts of FDI on energy efficiency are different between industry categories. Although some estimated coefficients are not significant, their signs show that FDI boosts mining industry's energy efficiency but reduces that of light and heavy industries. One possible reason is that, in mining industry FDI could help introduce advanced manufacturing techniques, management experience and working process from abroad. Meanwhile, foreign companies make FDI in China for the affluent and cheap labor force and mineral resources of China, and both light industry and heavy industry are labor-intensive and resource-intensive. As a consequence, we should treat FDI differently for different industries and advocate technology import so as to enhance the overall industrial energy efficiency in China.

4.2.6. Impacts of other influencing factors

With regard to labor-capital structure, except for mining industry, increase in capital-labor ratio will worsen energy efficiency, especially in pollution-intensive industries. Since the reform and opening-up, the capital-labor ratio in China increased and the industry has been experiencing a more heavy-oriented trend, which further impedes industrial energy efficiency from improving. As for technical progress, we use R&D investment and R&D researchers as proxy for technological investment and labor quality respectively. Generally, they both should be positively related with energy efficiency; however, the estimated coefficients are not significant. One possible explanation is the diversification of expenditures on technological investment, as we cannot peel off the parts relevant to energy efficiency improvement. In addition, in recent years, the government attaches much more importance to energy utilization efficiency and high-tech human capital nurturing, but it will take time for these efforts on energy efficiency improvement to take effect gradually.

5. Conclusion

With the improved Super-SBM model considering undesirable outputs, in this paper we study the energy efficiency and its development tendency of various industrial sectors in China as well as the four major industrial categories during the period from 2001 to 2010. Furthermore, we compare the energy efficiency difference between the "The Tenth Five-year Plan" and "The Eleventh Five-year Plan", and then explore the influencing factors of energy efficiency. The empirical results show that energy efficiency of each industrial sector and category during 2001–2010 has improved substantially. Particularly, during "The Eleventh Five-year Plan", energy efficiency of various industries has different degrees of improvement. However, efficiency variations across various industries have not shown a convergence trend. Energy efficiency of light industry is the highest among the four categories, followed by heavy industry. But the heavy industry has a higher growth rate compared with light industry, so the energy efficiency gap between these two industries has been reduced. Energy efficiency variations indicate prominent transformation in industrial economics.

For the influencing factors of energy efficiency, enterprise scale, industry concentration, industrial property rights structure, and government regulation have significant impacts on energy efficiency, and these factors lead to distinctive effects. Enterprise scale is an important determinant of energy efficiency. With expansion of enterprise scale, energy efficiency of various industries is reflected in different U-type curves. Industry concentration is positively related to energy efficiency, implying that high industry concentration benefits the enhancement of energy efficiency. Further reform of industrial state-owned enterprises will improve industrial energy efficiency. Moreover, increase in FDI and adjustment in energy consumption structure will improve industrial energy efficiency, especially when electricity consumption ratio is increased. Therefore, through adjusting the property rights structure, strengthening the technological and administrative spillover effects of FDI and developing clear, efficient new energy and substitute energy, the industrial energy efficiency will be enhanced to a great extent. More importantly, the above influencing factors and relevant policy advice for industrial energy efficiency are raised at the level of the entire industry as a whole, so the specific industrial groups and categories need different treatments. We cannot blindly stipulate a uniform energy-saving and emission-cutting target to enhance industrial energy efficiency. We should analyze the attributions and development situations of various industries, and deal with the influencing factors and regulation methods differently. Meanwhile, it is required to pay close attention to the relationship between improvements of energy efficiency and industrial developments, and to adjust this relationship in time.

Acknowledgments

This topic is a stage research outcome of the United States Energy Foundation project "Energy subsidy reform and China's sustainable economic development" in 2011 (Project No.: G-1111-15134); Ministry of Education, Philosophy, Social Planning project "Construct renewable energy industry finance risk management and policy support system based on the life cycle theory" in 2012 (Project No.: 12YJAZH056); China Postdoctoral Science Foundation "Sustainable development and social equity: a research based on the theory of energy subsidies and policy practice" in 2009 (Project No.20090460202); Special Funds for Construction Disciplines in Shanxi Province (201205737) and "Returning Students Scientific Research Project" of Shanxi Province (2012-023). We appreciate the comments of anonymous reviewers.

References

Andrews-Speed, P., 2009. China's ongoing energy efficiency drive: origins, progress and prospects. Energy Policy 37, 1331–1344.

Ang, B.W., Liu, F.L., Chung, H.S., 2004. A generalized Fisher index approach to energy decomposition analysis. Energy Econ. 26, 757–763.

Bunse, K., Vodicka, M., Schönsleben, P., Brülhart, M., Ernst, F.O., 2011. Integrating energy efficiency performance in production management – gap analysis between industrial needs and scientific literature. J. Clean. Prod. 19, 667–679.
Chen, T.Y., Yeh, T.L., 1998. A study of efficiency evaluation in Taiwan's banks. Int. J. Ser. Ind. Manag. 9, 402–415.
DeCanio, S.J., 1993. Barriers within firms to energy-efficient investments. Energy Policy 21, 906–914.
Färe, R., Grosskopf, S., 2004. Modeling undesirable factors in efficiency evaluation: comment. Eur. J. Oper. Res. 157, 242–245.
Hamilton, J.D., 1983. Oil and the macro economy since World War II. J. Polit. Econ. 91 (2), 228–248.
Hasanbeigi, A., Price, L., Lu, H., Lan, W., 2010. Analysis of energy-efficiency opportunities for the cement industry in Shandong Province, China: a case study of 16 cement plants. Energy 35, 3461–3473.
Hu, J.L., Wang, S.C., 2006. Total-factor energy efficiency of regions in China. Energy Policy 34, 3206–3217.
Hu, Y., 2012. Energy conservation assessment of fixed-asset investment projects: an attempt to improve energy efficiency in China. Energy Policy 43, 327–334.
Jenne, C.A., Cattell, R.K., 1983. Structural change and energy efficiency in industry. Energy Econ. 5, 114–123.
Lardic, S., Mignon, V., 2008. Oil prices and economic activity: an asymmetric cointegration approach. Energy Econ. 30 (3), 847–855.
Oda, J., Akimoto, K., Sano, F., Tomoda, T., 2007. Diffusion of energy efficient technologies and CO_2 emission reductions in iron and steel sector. Energy Econ. 29, 868–888.
Patterson, M.G., 1996. What is energy efficiency?: concepts, indicators and methodological issues. Energy Policy 24, 377–390.
Perc, M., Gómez-Gardeñes, J., Szolnoki, A., Floría, L.M., Moreno, Y., 2013. Evolutionary dynamics of group interactions on structured populations: a review. J. R. Soc. Interf. 10, 20120997. http://dx.doi.org/10.1098/rsif.2012.0997.
Perc, M., Szolnoki, A., 2010. Coevolutionary games—a mini review. BioSystems 99, 109–125.
Phylipsen, G., Blok, K., Worrell, E., 1997. International comparisons of energy efficiency methodologies for the manufacturing industry. Energy Policy 25, 715–725.
Ray, S.C., 2004. Data Envelopment Analysis: Theory and Techniques for Economics and Operations Research. Cambridge University Press.
Seiford, L.M., Zhu, J., 2002. Modeling undesirable factors in efficiency evaluation. Eur. J. Oper. Res. 142, 16–20.
Song, M.L., Wu, J., Wang, Y.M., 2012. Reducing energy demand in China: a statistical analysis of urban energy consumption in Anhui Province. Energy Environ. 23, 17–32.
Song, M.L., Zhang, L.L., An, Q.X., Wang, Z.Y., Li, Z., 2013. Statistical analysis and combination forecasting of environmental efficiency and its influential factors since China entered the WTO: 2002–2010–2012. J. Clean. Prod. 42, 42–51.
Sun, J.W., 1998. Changes in energy consumption and energy intensity: a complete decomposition model. Energy Econ. 20, 85–100.
Tone, K., 2001. A slacks-based measure of efficiency in data envelopment analysis. Eur. J. Oper. Res. 130, 498–509.
Tone, K., 2002. A slacks-based measure of super-efficiency in data envelopment analysis. Eur. J. Oper. Res. 143, 32–41.
Tone, K., 2003. Dealing with Undesirable Outputs in DEA: a Slacks-based Measure (SBM) Approach. In: GRIPS Research Report Series, pp. 1–5.
Tone, K., Sahoo, B.K., 2003. Scale, indivisibilities and production function in data envelopment analysis. Int. J. Prod. Econ. 84, 165–192.
Wang, Z.H., Zeng, H.L., Wei, Y.M., Zhang, Y.X., 2012. Regional total factor energy efficiency: an empirical analysis of industrial sector in China. Appl. Energy 97, 115–123.
Yang, W., Han, A., Wang, S.Y., 2013. Analysis of the interaction between crude oil price and US stock market based on interval data. Int. J. Energy Stat. 1 (2), 85–98.
Zhou, N., Levine, M.D., Price, L., 2010. Overview of current energy-efficiency policies in China. Energy Policy 38, 6439–6452.